

Last Time

Hypothesis Testing: Single Means
Research vs. Statistical Questions
The Null Hypothesis
The p -value

The Z-test

Test of Proportions

Today

Test of Proportions

T -tests of One Mean

Estimating σ^2

Test of Proportions

Let $\{X_1, X_2, \dots, X_N\}$ be a sequence of $(0,1)$ Bernoulli(p) observations, where p is the population probability of success. Then $\bar{X} = \hat{p}$, the proportion of “successes” in the sample and an estimate of p .

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We know that

$$\begin{aligned} E(X) = \mu &= p \text{ and} \\ \text{Var}(X) = \sigma^2 &= p(1 - p). \end{aligned}$$

So $E(\bar{X}) = E(X) = p$ and $\text{Var}(\bar{X}) = \sigma^2/N = p(1 - p)/N$.

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Central Limit Theorem: for large enough N , $\bar{X} = \hat{p} \sim \mathcal{N}(p, p(1 - p)/N)$.

This is one of those rare circumstances where we know σ^2 : the null hypothesis $H_0 : p = p_0$ determines both the mean and the variance of the sampling distribution of $\bar{X} = \hat{p}$.

Birth Order Effects

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$$H_1 : p > 0.5$$

Set $\alpha = 0.05$.

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Set $\alpha = 0.05$.

$$\bar{X} = 10/15 = 0.6667$$

$$\sigma_{\bar{X}}^2 = 0.5(1 - 0.5)/15 = 0.0167$$

$$Z_{\bar{X}} = \frac{0.6667 - 0.50}{\sqrt{0.0167}} = 1.2910$$

$$Pr(Z \geq 1.2910) = 0.0984$$

Family	Sibling		
	Older	Younger	
1	14	32	-
2	59	85	-
3	89	98	-
4	88	08	+
5	47	20	+
6	98	94	+
7	36	78	-
8	84	14	+
9	55	38	+
10	10	81	-
11	26	20	+
12	38	27	+
13	51	35	+
14	81	32	+
15	99	76	+

Birth Order Effects

```
prop.test(10,15,p=.5,alternative="greater")
```

1-sample proportions test with continuity correction

data: 10 out of 15, null probability 0.5

X-squared = 1.0667, df = 1, p-value = 0.1508

alternative hypothesis: true p is greater than 0.5

95 percent confidence interval:

0.4234848 1.0000000

sample estimates:

p

0.6666667

Remember the Correction For Continuity

Compare:

```
prop.test(10,15,p=.5,alternative="greater",correct=FALSE)
```

1-sample proportions test without continuity correction

data: 10 out of 15, null probability 0.5

X-squared = 1.6667, df = 1, p-value = 0.09835

alternative hypothesis: true p is greater than 0.5

95 percent confidence interval:

0.4551721 1.0000000

sample estimates:

p

0.6666667

Yates Correction

We computed $Pr(\hat{p} > 0.6667 \mid p_0 = .5) = Pr(Z > 1.2910) = 0.0984$.

But $\hat{p} = 0.6667$ was computed as $\frac{10}{15}$. Because $\bar{X} = \hat{p}$ is discrete,

$$Pr(\hat{p} \geq 10/15 \mid p_0 = 0.5) = Pr(\hat{p} > 9.5/15 \mid p_0 = 0.5).$$

The *real lower limit* of 10 is 9.5, which is where we need to consider the starting point to compute the area under the (continuous) normal density function.

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The *real lower limit* of 10 is 9.5, which is where we need to consider the starting point to compute the area under the (continuous) normal density function.

Recompute:

$$\hat{p} = 9.5/15 = 0.6333, \text{ so}$$

$$Z_{\hat{p}} = \frac{0.6333 - 0.5}{\sqrt{0.0167}} = 1.0318$$

$$Pr(Z \geq 1.0318) = 0.1508$$

Exact Binomial Test

The p -value with the Yates correction is 0.1508.

Compare this to the exact binomial test:

$$Pr(X \geq 10 \mid p = 0.5) = 0.1509$$

```
binom.test(10,15,alternative="greater")
```

```
Exact binomial test
```

```
data: 10 and 15
```

```
number of successes = 10, number of trials = 15, p-value = 0.1509
```

```
alternative hypothesis: true probability of success is greater than 0.5
```

```
95 percent confidence interval:
```

```
0.4225563 1.0000000
```

```
sample estimates:
```

```
probability of success
```

```
0.6666667
```

The t -distribution

t -table

```
bignum <- 10000 # number of simulated experiments
N <- 15 # sample size
pop.var <- 93 # population variance
pop.mean <- 47 # population mean

s2 <- S2 <- t.stat <- x.bar <- vector()

# Sample from a normal(pop.mean,pop.variance) distribution bignum times
# Compute the sample mean and variances and t-statistic
for (i in 1:bignum) {
  X <- rnorm(N,pop.mean,sqrt(pop.var))
  x.bar[i] <- mean(X)
  s2[i] <- var(X)
  S2[i] <- sum((X-x.bar[i])^2)/N # "Big S" squared, "population" variance
  t.stat[i] <- (x.bar[i] - pop.mean)/sqrt(s2[i]/N)
}

# What is the mean of the means?
cat("The population mean is ",pop.mean,
    " and the mean of the means is ",mean(x.bar))

# What is the mean of the variances?
cat("The population variance is ",pop.var,
    " and the mean of the sample variances is ",mean(s2))

# What is the mean of the "population" variances
cat("The population variance is ",pop.var,
    " and the mean of the "population" variances is ",mean(S2))
```

The t -distribution

```
> # What is the mean of  
the means?
```

```
The population mean is 47  
and the mean of the means is  
46.95188
```

```
>
```

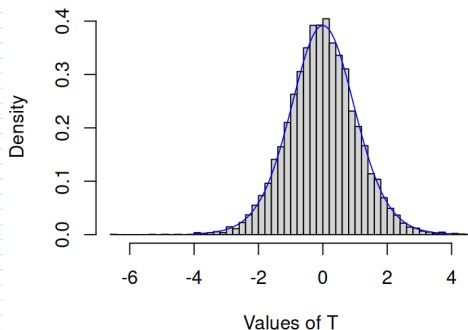
```
> # What is the mean of  
the variances?
```

```
The population variance is  
93 and the mean of the sample  
variances is 92.42257
```

```
>
```

```
> # What is the mean of  
the "population" variances
```

```
The population variance is  
93 and the mean of the  
'population' variances is  
86.26107
```



The Sample Variance

Given a sample $\{X_1, X_2, \dots, X_N\}$, we don't know σ^2 , but we can compute

$$s^2 = \frac{1}{N-1} \sum_{i=1}^N (X_i - \bar{X})^2.$$

We showed that s^2 a good estimate of σ^2 by simulation.

$E[s^2] = \mu_{s^2} = \sigma^2 \Rightarrow s^2$ is *unbiased*, and therefore a good point estimate of σ^2 is s^2 .

This means that $\hat{\sigma}_{\bar{X}} = s_{\bar{X}} = \frac{s}{\sqrt{N}}$, is a good estimate of the standard error of $\bar{X}$.

The (One-Sample) t -Test

Define: Test statistic

$$T_{\bar{X}} = \frac{\bar{X} - \mu}{s_{\bar{X}}},$$

where $s_{\bar{X}}$ is an estimate of $\sigma_{\bar{X}}$.

Two random variables (remember that any statistic is a random variable)

- ▶ $\bar{X} \sim \mathcal{N}(\mu, \sigma^2/N)$
- ▶ $s_{\bar{X}} \sim \text{something else}$
- ▶ $T_{\bar{X}} \sim t(N-1)$, kind of like a z-score but not quite.

GSS Example

Is the age of the GSS respondents significantly older than 30?

- One-tailed test:

$$H_0 : \mu \leq 30$$

$$H_1 : \mu > 30$$

- Test the null hypothesis at with a Type I error rate α equal to 0.05. (Always pick your α level before you do the test.)

GSS Example



$$\begin{aligned}\bar{X} &= 39.2462 \\ s^2 &= 196.1881 \\ N &= 873\end{aligned}$$

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$$T = \frac{\bar{X} - \mu_0}{s/\sqrt{N}} = \frac{39.2462 - 30}{14.0067/29.5466} = 19.5046$$

- ▶ $T \sim t(872) \approx \mathcal{N}(0, 1)$
- ▶ $Pr(T > 19.5046) = 0.0$ (p-value < 0.05)

T-tests in R

```
> t.test(data$AGE, alternative="greater", mu=30)
```

One Sample t-test

```
data: data$AGE  
t = 19.505, df = 872, p-value < 2.2e-16  
alternative hypothesis: true mean is greater than 30  
95 percent confidence interval:  
 38.4657      Inf  
sample estimates:  
mean of x  
 39.24628
```

Don't Forget

Confidence intervals around a single mean:

$$\bar{X} \pm t_{\alpha/2}(N-1) \frac{s}{\sqrt{N}}$$

What Can We Say About A Small p -Value?

Suppose $H_0 : \mu = 38$.

Test 1: $\bar{X} = 38.97$, $p = 0.02$

Test 2: $\bar{X} = 39.36$, $p = 0.002$ “more significant”

Rejecting $H_0 : \mu = 38$ says nothing about the alternative mean. Let the “true” alternative mean be $\mu_1 = 39$. Then the “true” sampling distribution of $\bar{X} \sim \mathcal{N}(39, \sigma_{\bar{X}})$, where $\hat{\sigma}_{\bar{X}} = 0.4741$.

